

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a8e3b0bf8f5ac8ed6.jsonl`  
- SHA-256: `be18e1d3947739f7dad07f7ee57d5911cd77649a1e2daf49e5972574f79e789`  
- Source modified: `2026-05-31T04:23:39+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:22:14.345Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Surya's model WEIGHTS are NOT permissively licensed: they use a modified AI Pubs OpenRAIL-M license that is free only for research, personal use, and organizations under \$5M funding/revenue, with a commercial license required above that threshold. This fails muneem's requirement that model weights be MIT/BSD/Apache-2.0 for unrestricted commercial use."

****Source:**** <https://github.com/VikParuchuri/surya/blob/master/LICENSE> (primary)

****Supporting quote:**** "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:22:16.959Z)

I'll verify this claim about Surya's model weights license. Let me start by checking the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-05-31T04:22:21.323Z)

Let me fetch the primary source (the Surya LICENSE file) and search for corroborating evidence in parallel.

4. user (2026-05-31T04:22:26.057Z)

Based on the content provided, this file specifies the **Apache License 2.0** for the code.

However, the document shown contains only the Apache License text itself. There is **no** information about model weights licensing in the provided content. The page displays the standard Apache 2.0 terms and conditions but does not mention:

- Model weights licensing
- OpenRAIL-M
- AI Pubs
- Commercial use restrictions
- Revenue/funding thresholds
- Research-specific terms

To find information about model weights licensing, you would need to check other files in the repository (such as a README, NOTICE file, or separate model license document).

5. user (2026-05-31T04:22:29.397Z)

Summary of Apache License 2.0 Content

The provided text is the **Apache License Version 2.0** from January 2004. Based on your request, I must note that this document contains **no** mentions of:

- Model weights
- AI Pubs
- OpenRAIL-M
- Commercial use restrictions
- Research use limitations
- Personal use provisions
- Startup exemptions
- Revenue or funding thresholds (such as \$5M)

What This License Actually Covers

This is a permissive open source software license that grants:

- Perpetual, worldwide, royalty-free rights to reproduce and create derivative works
- Patent licenses for covered contributions
- Ability to redistribute in Source or Object form

Key Conditions

The main requirements are:

1. Provide a copy of the license to recipients
2. Mark modifications clearly
3. Retain copyright and attribution notices
4. Include NOTICE files when applicable

The license explicitly disclaims warranties and limits liability, operating on an "AS IS" basis.

If you're looking for AI model licensing terms like those you mentioned, you would need to review a different license document (such as BigCode's OpenRAIL licenses, which specifically address model weights and usage restrictions).

6. user (2026-05-31T04:22:32.208Z)

Web search results for query: "Surya OCR VikParuchuri model weights license OpenRAIL-M commercial \$5M"

Links: [{"title": "surya-ocr · PyPI", "url": "https://pypi.org/project/surya-ocr/"}, {"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub", "url": "https://github.com/datalab-to/surya"}, {"title": "datalab-to/surya-ocr-2 · Hugging"}]

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Face", "url": "https://huggingface.co/datalab-to/surya-ocr-2"}, {"title": "GitHub -  
CibinQuadance/surya-OCR: OCR and line detection in 90+ languages ·  
GitHub", "url": "https://github.com/CibinQuadance/surya-OCR"}, {"title": "GitHub -  
cognitiveailab/surya-sci: OCR, layout analysis, reading order, line detection in 90+ languages ·  
GitHub", "url": "https://github.com/cognitiveailab/surya-sci"}, {"title": "surya/MODEL_LICENSE at  
master · datalab-to/surya", "url": "https://github.com/datalab-  
to/surya/blob/master/MODEL_LICENSE"}, {"title": "GitHub - kunthet/surya-text-line-detection-ocr:  
Accurate line-level text detection and recognition (OCR) in any language ·  
GitHub", "url": "https://github.com/kunthet/surya-text-line-detection-ocr"}, {"title": "GitHub -  
VikParuchuri/surya: OCR, layout analysis, reading order, table recognition in 90+  
languages", "url": "https://github.com/VikParuchuri/surya?tab=readme-ov-file"}, {"title": "Chandra  
OCR: The Open-Source AI That Actually Reads Tables and Handwriting  
Right", "url": "https://emelia.io/hub/chandra-ocr-tables-handwriting"}}
```

Based on the search results, here's the information about Surya OCR's licensing:

The Surya code is licensed under Apache 2.0, while the model weights use a modified AI Pubs Open Rail-M license that is free for research, personal use, and startups under \$5M funding/revenue. For broader commercial licensing of the model weights, there is a pricing page available.

The project was developed by Vik Paruchuri and is hosted on GitHub at the datalab-to organization.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-05-31T04:22:36.667Z)

Important finding: the LICENSE file at the repo root is just Apache 2.0 (covers the code). The model weights license is in a separate `MODEL\_LICENSE` file. The claim cites the wrong file path for the weights claim, but the substance (modified AI Pubs OpenRAIL-M, \$5M threshold) appears corroborated by the search. Also note the repo has moved from `VikParuchuri/surya` to `datalab-to/surya`. Let me fetch the actual MODEL_LICENSE and the README to verify the specifics.

8. user (2026-05-31T04:22:42.837Z)

AI PUBS OpenRAIL-M License Summary

****License Type:**** Yes, this is an OpenRAIL-M (Modified) license, Version 0.1, from March 2, 2023.

****Free Use Conditions:****

The license permits free use for personal use and research purposes without revenue/funding restrictions. However, commercial use faces limitations based on organizational size.

****Exact Dollar Thresholds:****

The license specifies two separate \$5,000,000 thresholds:

1. Revenue threshold: "for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars (\$5,000,000) in gross revenue in the prior year, except where Your Use is limited to personal use or research purposes"
2. Funding threshold: "for any purpose if You (your employer, or the entity you are affiliated with) has raised more than two million US dollars (\$5,000,000) in total equity or debt funding from any source, except where Your Use is limited to personal use or research purposes"

****Key Provision:****

Additionally, organizations cannot use the model "if You (your employer, or the entity you are affiliated with) provides or otherwise makes any product or service that competes with any product or service offered by or made available by Licensor or any of its affiliates."

Organizations exceeding these thresholds may obtain "commercial and broader use licenses" from the licensor.

9. user (2026-05-31T04:22:43.923Z)

Surya Licensing Summary

****Code License:****

"Code License Apache--2.0" - The Surya codebase is open source under Apache 2.0.

****Model Weights License:****

"Model License OpenRAIL--M" - The model weights use a modified AI Pubs Open Rail-M license.

****Usage Restrictions:****

The modified Open Rail-M license permits use for "research, personal use, and startups under \$5M funding/revenue." Beyond these categories, broader commercial licensing of model weights requires visiting their pricing page.

****Commercial Path:****

Organizations exceeding the \$5M threshold or seeking commercial deployment should visit Datalab's pricing page for licensing terms.

10. user (2026-05-31T04:22:48.048Z)

Web search results for query: "Surya OCR license commercial use weights "5 million" OR "\$5M" research only restriction"

Links: [{"title":"GitHub - CibiQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub","url":"https://github.com/CibiQuadance/surya-OCR"}, {"title":"GitHub - datalab-to/surya-OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub","url":"https://github.com/datalab-to/surya"}, {"title":"GitHub - cognitiveailab/surya-sci: OCR, layout analysis, reading order, line detection in 90+ languages · GitHub","url":"https://github.com/cognitiveailab/surya-sci"}, {"title":"surya-ocr · PyPI","url":"https://pypi.org/project/surya-ocr/"}, {"title":"pitapo/surya · Hugging Face","url":"https://huggingface.co/pitapo/surya"}, {"title":"View as Markdown","url":"https://replicate.com/cudanexus/ocr-surya/llms.txt"}, {"title":"Surya: The Revolutionary OCR Toolkit for 90+ Languages","url":"https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"}, {"title":"surya-ocr-vlite · PyPI","url":"https://pypi.org/project/surya-ocr-vlite/"}, {"title":"cudanexus/ocr-surya | Run with an API on Replicate","url":"https://replicate.com/cudanexus/ocr-surya"}, {"title":"datalab-to/surya-ocr-2 · Hugging Face","url":"https://huggingface.co/datalab-to/surya-ocr-2"}]

Based on the search results, here's what I found regarding Surya OCR's licensing terms for commercial use:

License Overview

The Surya code is licensed under Apache 2.0. The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).

Commercial Use Restrictions

The weights are licensed cc-by-nc-sa-4.0, but the author will waive that for any organization under \$5M USD in gross revenue in the most recent 12-month period.

Research and personal usage is always okay, but there are some restrictions on commercial usage.

For Organizations Over the \$5M Threshold

If you want to remove the GPL license requirements for inference or use the weights commercially over the revenue limit, you can contact the creator for dual licensing.

In summary, the model weights have a built-in restriction limiting free commercial use to organizations under \$5 million USD in gross revenue, while research and personal use are always permitted. Organizations exceeding this threshold need to obtain a commercial license.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-05-31T04:22:52.628Z)

I now have strong primary-source corroboration. Let me note one minor discrepancy in the WebFetch summary (it said "\$5,000,000" but quoted "two million US Dollars" inconsistently ? this is the small model garbling the parenthetical, not a substantive issue). The MODEL_LICENSE file itself is the authoritative AI Pubs OpenRAIL-M document. Let me do one direct read of the raw MODEL_LICENSE to nail down the exact threshold figure and confirm there's no misquote.

12. user (2026-05-31T04:22:59.116Z)

Commercial Use Thresholds

****Gross Revenue Threshold:****

"for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars (\$5,000,000) in gross revenue in the prior year"

****Funding Threshold:****

"for any purpose if You (your employer, or the entity you are affiliated with) has raised more than two million US dollars (\$5,000,000) in total equity or debt funding"

The verbatim numeric amounts are: ****two million US Dollars (\$5,000,000)**** in both clauses.

13. assistant (2026-05-31T04:23:05.170Z)

The small WebFetch model is producing a contradictory rendering ("two million US Dollars (\$5,000,000)") ? the word form and numeral disagree. This is a known failure mode of the summarizer model, not the source. The original AI Pubs OpenRAIL-M template uses "\$1,000,000" but Surya/DataLab's modified version is widely reported as \$5M. Let me get the raw bytes directly to resolve the exact figure definitively, and check a second independent source.

14. user (2026-05-31T04:23:06.899Z)

(a) for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars (\$5,000,000) in gross revenue in the prior year, except where Your Use is limited to personal use or research purposes;

(b) for any purpose if You (your employer, or the entity you are affiliated with) has raised more than two million US dollars (\$5,000,000) in total equity or debt funding from any source, except where Your Use is limited to personal use or research purposes; or

15. user (2026-05-31T04:23:12.420Z)

Licensing Summary

****Code License:****

The Surya code is licensed under Apache 2.0, as stated: "The Surya code is licensed under Apache 2.0."

****Model Weights License:****

The model weights use a modified AI Pubs Open Rail-M license with specific restrictions:

"The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)."

****Commercial Licensing:****

For broader commercial use beyond the stated restrictions, "For broader commercial licensing of

the model weights, visit our pricing page." The company offers a managed platform with commercial options and notes that startups must be under the \$5M funding/revenue threshold to qualify for the free tier.

****Key Distinction:****

The code itself has permissive open-source licensing, while the pre-trained model weights carry more restrictive terms requiring commercial arrangements for most business applications exceeding the startup revenue cap.

16. assistant (2026-05-31T04:23:26.726Z)

The raw license file confirms it definitively. The MODEL_LICENSE itself contains an internal inconsistency in the boilerplate (word form "two million" vs. numeral "\$5,000,000") ? this is a drafting artifact in Datalab's own modified template, which is why both my tools surfaced it. But this does not affect the substantive claim. The operative facts are all confirmed by the primary source:

- Code = Apache 2.0; weights = modified AI Pubs OpenRAIL-M (separate `MODEL\_LICENSE` file).
- Free only for personal use, research, and orgs under the revenue/funding threshold (\$5M per every secondary source and per the numeral in the file).
- Commercial license required above the threshold ? NOT permissive (not MIT/BSD/Apache for the weights).

Now let me assess the claim against the checklist.

****Checklist evaluation:****

1. ****Supported by the quote?**** The supporting quote ("modified AI Pubs Open Rail-M license, free for research, personal use, and startups under \$5M") fully matches the primary source. The claim is well-supported, not an overreach.
2. ****Contradicting evidence?**** None found. Every independent source (PyPI, GitHub README badge, multiple search snippets) agrees. No credible source disputes it.
3. ****Source quality?**** Primary source verified directly via raw GitHub file (`MODEL\_LICENSE`) plus the README. Sufficient for the claim's strength.
4. ****Outdated?**** Current. The repo moved from `VikParuchuri/surya` to `datalab-to/surya`, but the license terms persist on the active master branch as of now. The claim's cited URL (`VikParuchuri/surya/blob/master/LICENSE`) is technically the wrong path (that file is Apache code license; weights are in `MODEL\_LICENSE`), but the substantive claim about weights licensing is correct.
5. ****Marketing/speculation?**** No ? it's a license-terms claim grounded in the legal text.

****Minor caveats (do not refute the core claim):****

- The cited URL points to the Apache code `LICENSE`, not the `MODEL\_LICENSE` where the OpenRAIL-M terms actually live. The claim's *citation path* is imprecise, but the *substance* is correct and verifiable in the same repo.
- The license file itself has an internal word/numeral inconsistency on the dollar amount, but \$5M is the consistently reported and numerically stated figure.

The core claim ? Surya weights are NOT permissive, use modified AI Pubs OpenRAIL-M, free only for research/personal/under-\$5M, and thus fail muneem's MIT/BSD/Apache-for-weights requirement ? is accurate and primary-source verified. This is a legitimate license flag for muneem.

The claim survives.

17. user (2026-05-31T04:23:35.205Z)

Structured output provided successfully

18. assistant (2026-05-31T04:23:39.774Z)

Claim verified as accurate (refuted=false, high confidence). The Surya model weights are non-permissive (modified AI Pubs OpenRAIL-M, free only for research/personal/under-\$5M orgs), confirmed directly from the primary `MODEL\_LICENSE` file in the active `datalab-to/surya` repo, with corroboration from the README license badges and PyPI. The only flaw was the claim's cited URL pointing at the Apache code `LICENSE` rather than the sibling `MODEL\_LICENSE` ? a citation-path imprecision, not a substantive error. This is a legitimate license flag: Surya's weights fail muneem's MIT/BSD/Apache-for-weights requirement.